

Fig. 2: Pin configurations of regulators

use heat-sinks for the regulators.

The three-terminal regulators are almost non-destructible. These have inbuilt protection circuits including the thermal shutdown protection. Even if there is overload or shorting of the output, the inbuilt overload protection circuit will limit the current and slowly reduce the output voltage to zero. Similarly, if the temperature increases beyond a certain value due to excessive load and heat dissipation, the in-built thermal shutdown circuit will reduce the output current and the output voltage (gradually) to zero. Thus complete protection is provided to the circuitry.

Assemble the circuit on a general-purpose PCB and enclose in a box as shown in Fig. 3.

The step-by-step procedure to build the triple power supply for the laboratory follows:

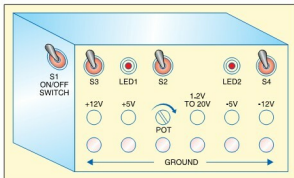


Fig. 3: Proposed cabinet for power supply

1. Collect all the components shown in the circuit diagram.
2. Connect switch S1, fuse, transformer and mains cord to the assembled PCB as well as the box.
3. Keep the multimeter in DC voltage range (more than 25V DC) and measure the DC voltage across capacitors C1 and C7 (1000 μ F, 35V). This voltage should be around $18V \times 1.41 = 25$ to 26V DC. Check both positive and negative voltages with respect to ground.
4. It is advisable to use three-wire mains cable and plug. If you are using any metallic box, earthing wire/pin of the mains plug should be soldered to the body of the metallic box using an

earthing tag.

5. If the 18V-0-18V transformer is replaced with 15V-0-15V transformer, the output voltage of the variable supply using LM317 will be correspondingly lower.

6. If proper voltages are available, go to step 7. Otherwise, check the connections.

7. Connect variable regulator LM317 to the circuit and check 1.25V to 20V output by varying the 2.2-kilo-ohm linear potentiometers.

8. Now connect ICs 7812, 7912, 7805 and 7905 to the circuit and check their output voltage.

9. Connect terminals, potmeter, switches and indicator LED on the front panel of the box and complete the connections. Close the box by using screws.

Precaution. At the primary side of the transformer, 230V AC could give lethal shocks. So be careful not to touch this part. EBY will not be responsible for any resulting loss or harm to the user. ●

UNIQUE WATER PUMP CONTROLLER



■ **D. MOHAN KUMAR**

Here is a simple solution for automatic pumping of water to the overhead tank. Unlike other water-level indicators, it does not use probes to detect the water level and hence there is no probe corrosion problem. It has no direct contact with water, so the chance of accidental leakage of electricity to the water tank is also eliminated. Two important advantages of the circuit are that the water

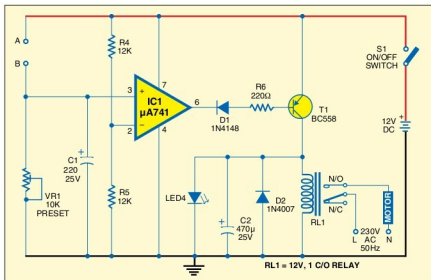


Fig. 1: Water pump controller